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S E 417
HW3
03/27/26

Question 1:

Hand in the first and last 5 test frames of this file and answer the following questions

Test Case 1 (Key = 1.1.1.1.1.1.1.)
help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_alphabetically
duplicates : yes

Test Case 2 (Key = 1.1.1.1.1.1.2.)
help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_alphabetically
duplicates : no

Test Case 3 (Key = 1.1.1.1.1.2.1.)
help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_numerically
duplicates : yes

Test Case 4 (Key = 1.1.1.1.1.2.2.)
help : on
version : on
order : numeric

check : yes
unique : yes
file_contents : sorted_numerically
duplicates : no

Test Case 5 (Key = 1.1.1.1.1.3.1.)
help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : unsorted
duplicates : yes

…

Test Case 380 (Key = 2.2.4.2.2.1.2.)
help : off
version : off
order : reverse
check : no
unique : no
file_contents : sorted_alphabetically
duplicates : no

Test Case 381 (Key = 2.2.4.2.2.2.1.)
help : off
version : off
order : reverse
check : no
unique : no
file_contents : sorted_numerically
duplicates : yes

Test Case 382 (Key = 2.2.4.2.2.2.2.)
help : off
version : off
order : reverse
check : no
unique : no

file_contents : sorted_numerically
duplicates : no

Test Case 383 (Key = 2.2.4.2.2.3.1.)
help : off
version : off
order : reverse
check : no
unique : no
file_contents : unsorted
duplicates : yes

Test Case 384 (Key = 2.2.4.2.2.3.2.)
help : off
version : off
order : reverse
check : no
unique : no
file_contents : unsorted
duplicates : no

(a) Pick one test frame and explain (on paper) what this frame means (i.e. what combinations of options you would test based on the frame – note you will want to choose one that does not have the help and/or the version turned on for this question). Use the existing input files to pick a file that would make sense to use for this test frame. Give this frame as a concrete test case and show a screen shot of the output (you will need to show the command line so we know which input you used).

Test Case 310 (Key = 2.2.1.2.2.2.2.)
help : off
version : off
order : numeric
check : no
unique : no
file_contents : sorted_numerically
duplicates : no

This test case ensures that the numeric sort works on a file that is already sorted numerically and has no duplicates. Essentially, this is a sanity check. check: no and unique: no means there is no -c or -u in the command.

```
nathan-krieger  main  sort -n sorted-numbers-unique.txt
3
4
10
90
101
345
455
900
```

(b) Now run the tsl on the file called “sort.refined.tsl”. How does sort.refined.tsl modify the tests cases that used help and version. Explain the differences and list how many frames are in the refined file

```
nathan-krieger  main  ./tsl sort.refined.tsl

      98 test frames generated and written to sort.refined.tsl.tsl
```

Sort refined tsl is different because it removes the redundant tests containing help/version. Since those two are meant to be exclusive (meaning if help is on the other flags don’t matter, if version is on, the other flags don’t matter) there is no need to test every combination with them. The sort refined tsl generated 98 test frames.

Question 2:

(a) How many combinations (test cases) are there for each of the above files?

./cit_generate input.txt -F

    9 combinations

./cit_generate cit-sort-input.txt -F

    12 combinations

./cit_generate cit-sort-input-3way.txt -F

    24 combinations

(b) Provide the output from the first one (cit-sort-input.txt)

```
nathan-krieger  main -  cat cit_sample.out
t   k   v   TCount
2 5 13 66

 4 2 2 3 2

  0  1  2  3
  4  5
  6  7
  8  9  10
  11  12

12
 1 4 7 8 12
 0 5 7 8 11
 3 5 6 10 12
 0 5 6 9 12
 1 5 6 9 11
 2 5 6 8 11
 0 4 7 10 11
 1 4 6 10 11
 2 4 7 9 12
 2 5 7 10 11
 3 5 7 9 11
 3 4 7 8 12
```

(c) (Write your own program) or manually map the test cases from cit-sort-input.txt so that they use the choices from the TSL.

Cit-sort-input.txt:

```
[~/git/SE417/HW3/CIT_Tool]
nathan-krieger  main  cat cit-sort-input.txt
2
5
4 1
2 2
3 1
2 1
```

Mapping:
Combination strength is 2
5 paramerters
The first param has 4 options
The next 2 have 2 options
The 4th parameter has 2 options
The 5th parameter has 1 option

	Switch category	Numbers	Mapping
4 1	order	0 1 2 3	0 = numeric 1 = default 2 = random 3 = reverse
2 2	check	4 5	4 = yes 5 = no
2 2	unique	6 7	6 = yes 7= no
3 1	File contents	8 9 10	8 = alpha 9 = numeric 10 = unsorted
2 1	duplicates	11 12	11 = yes 12 = no

The 12 test cases:
1 4 6 8 11
1 5 6 10 11
2 5 7 10 11

1 5 7 9 12
3 5 6 10 11
3 5 6 9 12
2 5 6 9 12
3 4 7 8 12
0 5 7 8 11
2 4 6 8 11
0 4 6 9 11
0 4 7 10 12

Mapped tests:

Line 1: 1 4 6 8 11
1: Order = default(alpha)
4: Check = yes
6: Unique = yes
8: File Contents = sorted_alphabetically
11: Duplicates = yes
Mapped Test: sort -c -u sorted.words.txt

Line 2: 1 5 6 10 11
1: Order = default(alpha)
5: Check = no
6: Unique = yes
10: File Contents = unsorted
11: Duplicates = yes
Mapped Test: sort -u random-numbers.txt

Line 3: 2 5 7 10 11
2: Order = random
5: Check = no
7: Unique = no
10: File Contents = unsorted
11: Duplicates = yes
Mapped Test: sort -R random-numbers.txt

Line 4: 1 5 7 9 12
1: Order = default(alpha)
5: Check = no
7: Unique = no
9: File Contents = sorted_numerically
12: Duplicates = no
Mapped Test: sort sorted-numbers-unique.txt

Line 5: 3 5 6 10 11
3: Order = reverse
5: Check = no
6: Unique = yes
10: File Contents = unsorted
11: Duplicates = yes
Mapped Test: sort -r -u random-numbers.txt

Line 6: 3 5 6 9 12
3: Order = reverse
5: Check = no
6: Unique = yes
9: File Contents = sorted_numerically
12: Duplicates = no
Mapped Test: sort -r -u sorted-numbers-unique.txt

Line 7: 2 5 6 9 12
2: Order = random
5: Check = no
6: Unique = yes
9: File Contents = sorted_numerically
12: Duplicates = no
Mapped Test: sort -R -u sorted-numbers-unique.txt

Line 8: 3 4 7 8 12
3: Order = reverse
4: Check = yes
7: Unique = no
8: File Contents = sorted_alphabetically
12: Duplicates = no
Mapped Test: sort -r -c sorted.words.txt

Line 9: 0 5 7 8 11
0: Order = numeric
5: Check = no
7: Unique = no
8: File Contents = sorted_alphabetically
11: Duplicates = yes
Mapped Test: sort -n sorted.words.txt

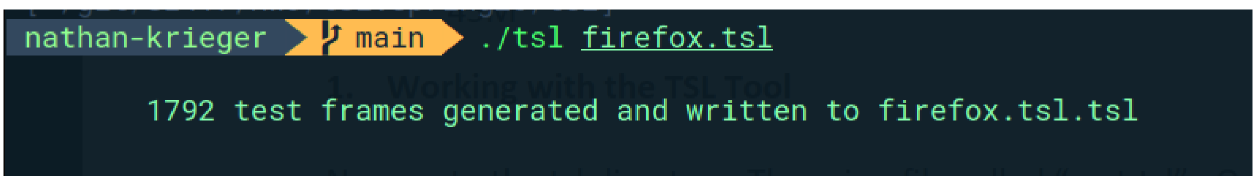
Line 10: 2 4 6 8 11
2: Order = random
4: Check = yes
6: Unique = yes

8: File Contents = sorted_alphabetically
11: Duplicates = yes
Mapped Test: sort -R -c -u sorted.words.txt

Line 11: 0 4 6 9 11
0: Order = numeric
4: Check = yes
6: Unique = yes
9: File Contents = sorted_numerically
11: Duplicates = yes
Mapped Test: sort -n -c -u sorted-numbers-dups.txt

Line 12: 0 4 7 10 12
0: Order = numeric
4: Check = yes
7: Unique = no
10: File Contents = unsorted
12: Duplicates = no
Mapped Test: sort -n -c random-numbers.unique.txt

Question 3:



(a) How many test frames do you have in this file?

1792

(b) Copy the contents of the firefox.tsl to the document. Copy the first and last 5 tests (10 in total) of the firefox.tsl.tsl file to this document

```
#tsl for sort
Parameters:
    search_engine:
      Google.
      Amazon.
      Bing.
      DuckDuckGo.
      eBay.
      Perplexity.
      Wikipedia.
    tab_opening_links_in_tabs:
        on.
        off.
    tab_opening_switch_to_link:
        on.
        off.
    tab_opening_links_next_to_active_tab:
        on.
        off.
    tab_interaction_ctrl_tab_recently_used:
        on.
        off.
    tab_interaction_image_preview:
        on.
        off.
    browser_layout:
      horizontal_tabs.
      vertical_tabs.
    firefox_suggest_history:
        on.
        off.
    firefox_suggest_bookmarks:
        on.
        off.
```

Test Case 1 (Key = 1.1.1.1.1.1.1.1.1.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs

firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 2 (Key = 1.1.1.1.1.1.1.1.2.)
search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : off

Test Case 3 (Key = 1.1.1.1.1.1.1.2.1.)
search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : on

Test Case 4 (Key = 1.1.1.1.1.1.1.2.2.)
search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Test Case 5 (Key = 1.1.1.1.1.1.2.1.1.)
search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on

tab_interaction_image_preview : on
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 1788 (Key = 7.2.2.2.2.2.1.2.2.)
search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Test Case 1789 (Key = 7.2.2.2.2.2.2.1.1.)
search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 1790 (Key = 7.2.2.2.2.2.2.1.2.)
search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : off

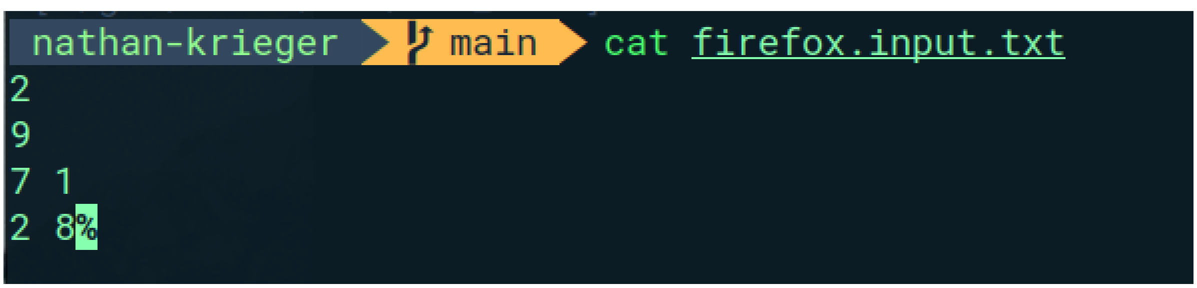
Test Case 1791 (Key = 7.2.2.2.2.2.2.2.1.)
search_engine : Wikipedia
tab_opening_links_in_tabs : off

tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : on

Test Case 1792 (Key = 7.2.2.2.2.2.2.2.2.)
search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Question 4:

(a) Hand in a screen shot of both of these files (the input file and the 2-way sample).



```
nathan-krieger  main  cat firefox.input.txt
2
9
7 1
2 8%
```

```
nathan-krieger  main  cat cit_sample.out
t   k   v  TCount
2 9 23 224

 7 2 2 2 2 2 2 2 2

  0  1  2  3  4  5  6
  7  8
  9  10
  11  12
  13  14
  15  16
  17  18
  19  20
  21  22

14
 6 8 9 12 13 16 18 19 21
 0 8 10 12 14 16 18 20 21
 4 7 9 11 14 15 18 20 22
 2 7 10 12 14 16 17 20 22
 4 8 10 12 13 16 17 19 21
 5 8 10 12 13 15 17 20 21
 3 8 10 11 14 16 17 20 22
 1 8 10 12 14 15 18 19 21
 6 7 10 11 14 15 17 20 22
 1 7 9 11 13 16 17 20 22
 2 8 9 11 13 15 18 19 21
 3 7 9 12 13 15 18 19 21
 5 7 9 11 14 16 18 19 22
 0 7 9 11 13 15 17 19 22
```

(b) Map the 2-way sample file to the configuration names and hand in the mapped file. This should match the file handed in in part a.

Integers	**Parameter**	**Mapping**
0 - 6	Search Engine	0=Google, 1=Amazon, 2=Bing, 3=DuckDuckGo, 4=eBay, 5=Perplexity, 6=Wikipedia
7, 8	Tab: Opening Links	7=on, 8=off
9, 10	Tab: Switch to Link	9=on, 10=off
11, 12	Tab: Next to Active	11=on, 12=off
13, 14	Tab: Ctrl+Tab	13=on, 14=off
15, 16	Tab: Image Preview	15=on, 16=off
17, 18	Browser Layout	17=Horizontal, 18=Vertical
19, 20	Suggest: History	19=on, 20=off
21, 22	Suggest: Bookmarks	21=on, 22=off

1. 6 8 9 12 13 16 18 19 21 Wikipedia, Tab-Links-Off, Switch-Link-On, Next-Active-Off, Ctrl-Tab-On, Preview-Off, Vertical-Layout, History-On, Bookmarks-On

2. 0 8 10 12 14 16 18 20 21 Google, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-Off, Vertical-Layout, History-Off, Bookmarks-On

3. 4 7 9 11 14 15 18 20 22 eBay, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-Off, Preview-On, Vertical-Layout, History-Off, Bookmarks-Off

4. 2 7 10 12 14 16 17 20 22 Bing, Tab-Links-On, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

5. 4 8 10 12 13 16 17 19 21 eBay, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-On, Preview-Off, Horizontal-Layout, History-On, Bookmarks-On

6. 5 8 10 12 13 15 17 20 21 Perplexity, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-On, Preview-On, Horizontal-Layout, History-Off, Bookmarks-On

7. 3 8 10 11 14 16 17 20 22 DuckDuckGo, Tab-Links-Off, Switch-Link-Off, Next-Active-On, Ctrl-Tab-Off, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

8. 1 8 10 12 14 15 18 19 21 Amazon, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-On, Vertical-Layout, History-On, Bookmarks-On

9. 6 7 10 11 14 15 17 20 22 Wikipedia, Tab-Links-On, Switch-Link-Off, Next-Active-On, Ctrl-Tab-Off, Preview-On, Horizontal-Layout, History-Off, Bookmarks-Off

10. 1 7 9 11 13 16 17 20 22 Amazon, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

11. 2 8 9 11 13 15 18 19 21 Bing, Tab-Links-Off, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-On, Vertical-Layout, History-On, Bookmarks-On

12. 3 7 9 12 13 15 18 19 21 DuckDuckGo, Tab-Links-On, Switch-Link-On, Next-Active-Off, Ctrl-Tab-On, Preview-On, Vertical-Layout, History-On, Bookmarks-On

13. 5 7 9 11 14 16 18 19 22 Perplexity, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-Off, Preview-Off, Vertical-Layout, History-On, Bookmarks-Off

14. 0 7 9 11 13 15 17 19 22 Google, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-On, Horizontal-Layout, History-On, Bookmarks-Off

(c) Assume you have a base choice using the default values above. You can assume that Bing is the default browser.

Create a sample for Base Choice Criteria (this has to be done by hand)

Hand in the Base Choice criteria

Base choice:

Search engine: Bing

All other settings: on

Test #	Search Engine	Tab: Links	Tab: Switch	Tab: Next	Tab: Ctrl+Tab	Tab: Preview	Layout	Sug: History	Sug: Bookmarks
1	**Bing**	**on**	**off**	**off**	**off**	**on**	**Horiz.**	**on**	**on**
2	Google	on	off	off	off	on	Horiz.	on	on
3	Amazon	on	off	off	off	on	Horiz.	on	on
4	Duck Duck Go	on	off	off	off	on	Horiz.	on	on
5	eBay	on	off	off	off	on	Horiz.	on	on
6	Perplexity	on	off	off	off	on	Horiz.	on	on
7	Wikipedia	on	off	off	off	on	Horiz.	on	on

8	Bing	**off**	off	off	off	on	Horiz.	on	on
9	Bing	on	**on**	off	off	on	Horiz.	on	on
10	Bing	on	off	**on**	off	on	Horiz.	on	on
11	Bing	on	off	off	**on**	on	Horiz.	on	on
12	Bing	on	off	off	off	**off**	Horiz.	on	on
13	Bing	on	off	off	off	on	**Vert.**	on	on
14	Bing	on	off	off	off	on	Horiz.	**off**	on
15	Bing	on	off	off	off	on	Horiz.	on	**off**